

Navigating Paris' Metro

Graph Theory Project Report by LESTERLIN Raphaël, DONNAT Arthur and DELANNOY Arthur

Context

The Floyd-Warshall algorithm, applied on a graph, allows after some lengthy calculations to obtain two matrices which give all possible minimum-cost paths between one vertex and another one in the whole graph. Graphs are often used to modelize in a quick and visually understandable way problems of transportation, be it through research operations, or in the case we will study axis of transit throughout a given geographical unit.

Paris has the 2nd most dense metropolitan network in the whole world [1], it is the perfect transportation system to apply this algorithm on. We will thus try to find the minimum cost path across the network to get from point A to point B.

Data

To keep a reasonable scope for this project, we immediately put some constraints on our attempt due to the massive number of stations and lines across the network. We decided to restrain the vertices to the set of stations providing a connection to other lines of metro, in the fourteen historical lines of the Parisian metro. Our cost will be modeled by the number of stations in-between a stop and another one, as we do not have the time to empirically check the average time for an ideal rolling stock to go towards the destination. Using the RATP website [2], we put in an excel sheet the list of connected stations for each line, identifying with a unique ID each station. Then, we used this data to create the graph as a list of arcs from one transit nexus to the next. The file containing it will be given as an appendix, since it would be too long to fit in this document in an appropriate manner.

Results

Running our algorithm on the graph, we obtain the following execution trace :

And the end result :

To test our program's accuracy, we determine the minimum cost path (the "shortest") and compare to the propositions of apps (Google Maps, City Mapper) to see if they are both coherent.

[insérer différents tests avec des stations assez loin des autres]

Conclusion

[insert if it works or not]

Bibliography

[1] [Île-de-France Mobilités : orchestrer la plus grande révolution des mobilités en Europe | Île-de-France Mobilités](#)

[2] [Plan des lignes du métro, RER, bus et tramway | RATP](#)